

HTIL x CFSG · RESEARCH PROPOSAL

The Next Interface Is Human

A brain-computer interface and robotics research partnership between the Human-Technology Interaction Lab (University of Alabama) and the Computing for Social Good Lab (University of Florida).

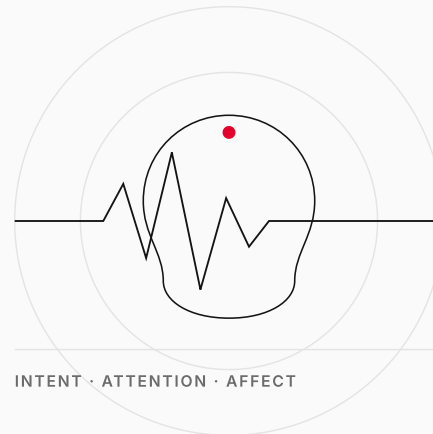
Prepared for The Coca-Cola Company — Dr. Chris S. Crawford · Dr. Juan E. Gilbert.

CONTENTS

THE OPENING QUESTION

Coca-Cola has engineered the drink. The human at the machine is still an open question.

The next frontier is not the formula — it is the brain and body at the moment of choice: intent, attention and affect, served through machines that move in the real world.



WHY NOW

Four frontiers, one shared problem: the human interface.

01 · PERSONALIZATION

Drinks personalized by AI. The next step adapts to the person's state, not just their history.

02 · CONNECTED DISPENSING

Freestyle machines keep getting smarter. The interface is still a screen we touch.

03 · INCLUSIVE ACCESS

Hands-free control opens the machine to users who cannot touch it at all.

04 · BRAND EXPERIENCE

Embodied hosts and measured engagement at live venues, instead of logo exposure.

Every one of these is a human-interface problem — and human interfaces are what we build.

THE PAIRING

Two labs, one shared specialty: the human as interface.

HTIL · UNIVERSITY OF ALABAMA

Dr. Chris S. Crawford

Brain-computer interfaces & human-robot interaction

Reads **attention, arousal and affect** in real time

NeuroBlock — low-code neurofeedback; beginners program with their own brain signals

Low-cost physiological sensing hardware, **~\$4 BOM**

NSF CAREER award

CFSG · UNIVERSITY OF FLORIDA

Dr. Juan E. Gilbert

Human-centered computing & embodied humans

National Medal of Technology and Innovation, 2023 · ACM Fellow

BCI among the lab's **core research areas**

Prime III — open-source universal design, used in real elections

Applications Quest — patented equitable AI

One lab reads the signal. Both labs turn it into an interface everyone can use.

The full brain-to-machine stack.

Neural & physiological sensing	EEG, EMG and affect, from low-cost hardware upward.
Signal decoding	Turning waveforms into intent, attention and engagement.
Human-robot interaction	Embodied agents that sense and respond to a person.
Low-code authoring & evaluation	NeuroBlock-style tooling plus rigorous study design.

From electrode to embodied experience, in one research program.

Six projects at the brain-machine boundary.

A • The Responsive Pour	Neuro-adaptive personalization
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B • The Hands-Free Pour	Accessible BCI control
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C • The Living Lab	Robot hosts at live venues
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D • The Experience Signal	Neurophysiological measurement
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E • NeuroBlock Studio	Low-code BCI & robot authoring
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F • The Serving Arm	Service & assistive robotics
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Each is a research question — designed to produce peer-reviewed results and protectable IP.

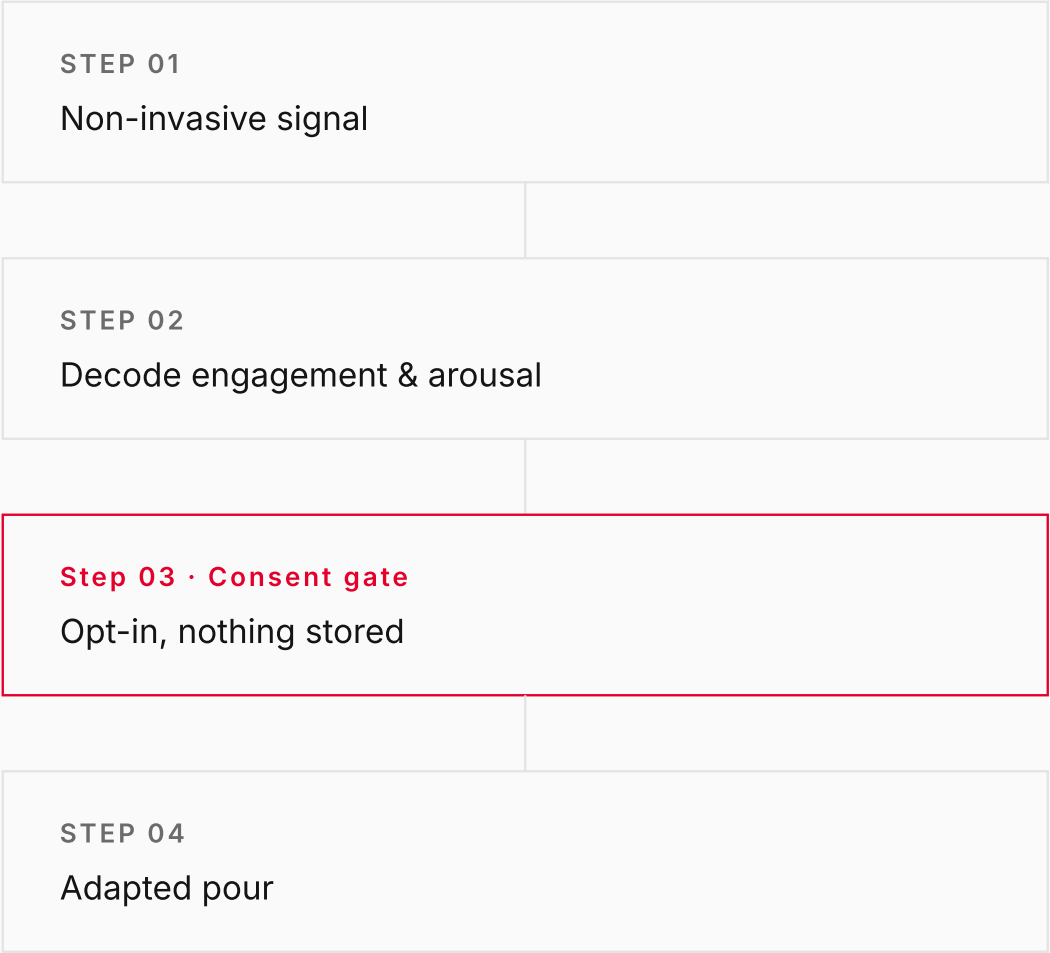
The responsive pour.

WHAT WE BUILD

Dispensing that reads engagement and arousal from non-invasive EEG and physiological signals — never identity — and adapts the Freestyle moment in real time.

WHY IT MATTERS TO COCA-COLA

Personalization that responds to how a person actually feels, without covert sensing or stored profiles.



The hands-free pour.

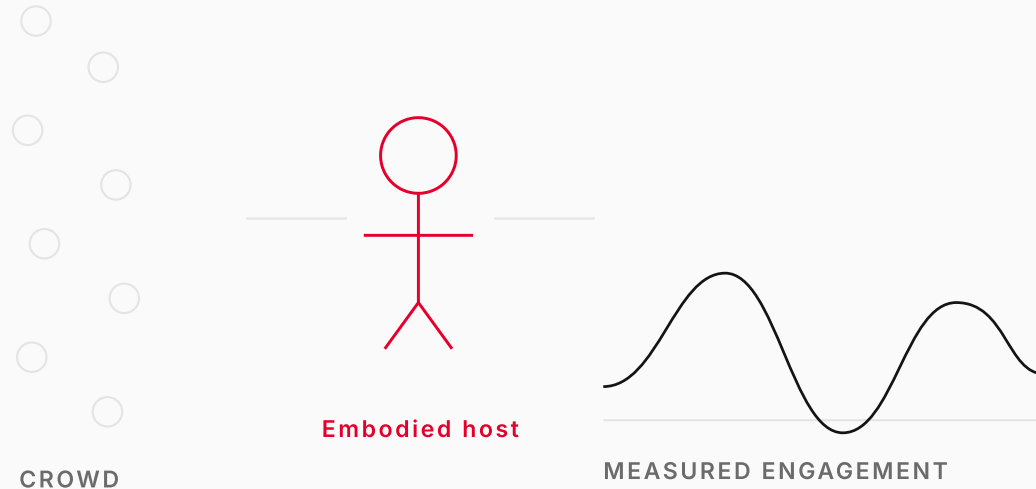
WHAT WE BUILD

A brain- and body-controlled interface — attention, gaze and physiological signals — that lets a user select and pour without touching the machine, built on the low-cost sensing stack and universal-design method.

WHY IT MATTERS TO COCA-COLA

Serves motor-impaired and aging customers, sterile and sanitation-critical settings, and anyone whose hands are full — and puts Coca-Cola at the front of accessible beverage control.

The living lab.



WHAT WE BUILD

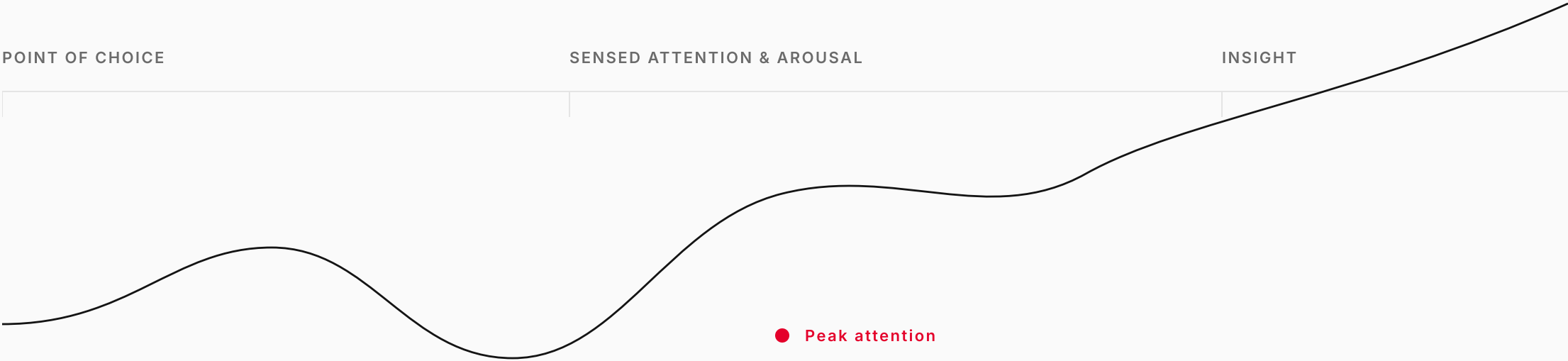
Embodied hosts — robotic and virtual — at sponsored venues, with physiological sensing to measure real engagement and test human-robot trust in hospitality.

WHY IT MATTERS TO COCA-COLA

Turns sponsorship from logo exposure into measured experience science that feeds back into products and venues.

The experience signal.

A rigorous, ethics-governed instrument measuring attention and engagement using EEG and physiological signals — extending neuromarketing from the lab to the real point of choice, and giving Coca-Cola an honest, independent read on what resonates.



NeuroBlock Studio.

WHAT WE BUILD

The block-based paradigm proven in NeuroBlock, rebuilt as an authoring toolkit so Coca-Cola's experiential and agency teams compose brain-driven and robot-driven experiences without a neuroscientist in the room.

WHY IT MATTERS TO COCA-COLA

Turns one-off activations into a reusable capability — faster ideation, lower cost, and a shared language between marketers and engineers.

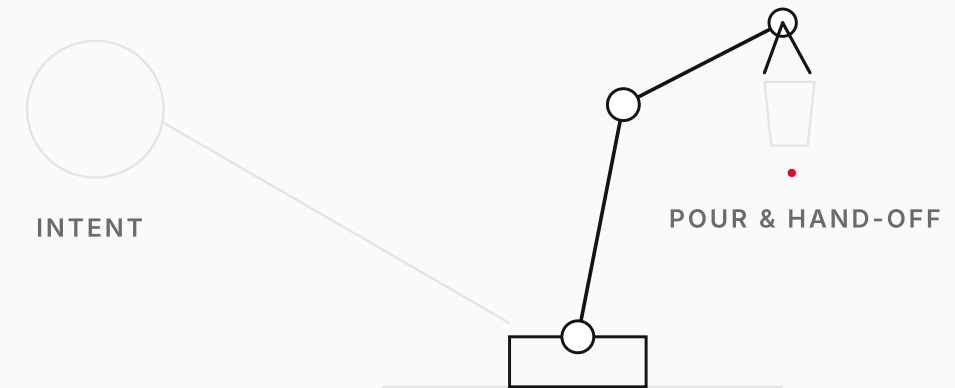
The serving arm.

WHAT WE BUILD

A BCI-driven robotic pour-and-hand-off system — assistive for users who cannot operate a dispenser, and a hospitality robot for high-traffic venues.

WHY IT MATTERS TO COCA-COLA

A credible entry into service robotics for beverage, with the human-factors evidence most robotics deployments skip.



TRUST & ACCEPTANCE READOUT

THE RETURN

Not a vendor. A research partner.

PUBLISHABLE, PATENTABLE IP

Peer-reviewed BCI and robotics results, co-owned under a negotiated agreement.

DE-RISKED INNOVATION

Small, evidence-backed pilots before any large commitment.

CREDIBILITY

Brain-interface and robotics claims backed by university research, with an ethics framework built in.

TALENT

A pipeline to trained graduate researchers in BCI, HRI and physiological computing.

THE MODEL

How we would work together.



STRUCTURE

Two PIs — Crawford and Gilbert — one program manager, four to six graduate researchers across both campuses.

FUNDING

Sponsored research matched by federal and foundation grants, lowering net cost per project.

DELIVERABLES

Quarterly findings, published papers, working BCI and robot prototypes, annual report to leadership.

The drink is already
engineered to a science. The
human at the machine is not
— yet.

HTIL × CFSG — a proposed BCI and robotics partnership with The
Coca-Cola Company.

THE ASK: A 90-DAY SCOPING SPRINT TO PICK THE FIRST TWO PROJECTS.